Why Our Knowledge Looks Like This: Representation, Path Dependence, and a Question from High School

In high school, I started asking a question that no one around me could really answer: why do we have to learn this? I did not only mean subjects I was not interested in. I meant something larger. Why does our knowledge look the way it does? Why are mathematical formulas written in this particular way, and why did our civilization develop these sciences and social structures rather than others? If the earliest civilizations had taken a different path, would we now be learning completely different things? I even imagined parallel universes in which history had gone another way. Even when a theorem is proved, the truth it describes was already there before anyone proved it. So why does that truth appear to us in this particular form? At the time, these questions felt like a trap. I still had to study for exams, and asking why only made studying harder.

Studying cognitive science has given me words for what I was asking. Theorems, formulas, and number systems are representations: forms in which knowledge is written down. Once created, they are used for computation, to solve problems, build tools, and produce new representations, which are used in turn. Each round builds on the one before. What I was really asking was whether this process had to arrive where it did.

In this paper, I use Marr's levels of analysis to separate two versions of that question. I argue that what our knowledge is about is constrained by the world, so different civilizations facing the same problems are likely to converge on similar content. But how that knowledge is represented is constrained from below, by the human mind and by earlier representations, and is therefore path-dependent. Our representations are not the only possible ones, but they are not arbitrary either, and the ones we inherit decide what is easy to compute next.

The Truth and Its Representation

Marr (1982) proposed that an information-processing system can be described at three levels: the computational level, which specifies what problem is solved and why; the algorithmic level, which specifies the representation of the input and output and the steps that transform one into the other; and the implementation level, which describes the physical system that carries out

those steps. My high-school question mixed the first two levels. What a theorem says — for example, that for any right triangle, the two shorter sides fix the longest side in the same way — is like the computational level: it specifies what holds, regardless of how anyone writes it. How the theorem is written and proved is like the algorithmic level: a choice of representation and of steps.

The Pythagorean theorem shows this clearly. For every right triangle, the square on the longest side equals the sum of the squares on the other two sides. This was true of every right triangle long before anyone stated it. Yet it has been represented in very different ways. In Euclid's *Elements*, it is proved step by step from axioms and earlier propositions (Euclid, ca. 300 B.C.E./1998). In Chinese mathematics, the same relationship appears as the *gougu* theorem in the *Zhoubi Suanjing*, named after the two sides of a carpenter's square, and a third-century commentary by Zhao Shuang includes a "hypotenuse diagram" that shows the result by arranging four copies of a right triangle inside a square and comparing areas (Cullen, 1996). One representation persuades through a chain of logical steps; the other persuades by letting you see that the pieces fit. The truth is the same. What differs is the path by which a human mind is brought to accept it.

Marr's own example was numbers. The same number can be written in Arabic numerals, Roman numerals, or binary. Thirteen is thirteen in all three, but each representation makes some information explicit and hides other information. Arabic numerals make the decomposition into powers of ten explicit; binary makes powers of two explicit instead. This matters for computation. Multiplying 37 by 24 in Arabic numerals is a routine procedure taught to children. Multiplying XXXVII by XXIV in writing is much harder, because Roman numerals do not line digits up by place value, so the column procedure has nothing to work on. A representation does not change what is true, but it changes what is easy.

In class, we described a representation as a model of the world that guides action. Different models of the same truth can guide action equally well in principle while making some actions much easier in practice. So the answer to "why does the truth appear in this form?" cannot come from the truth itself. A fact does not choose its own representation. Something else does, and that is where the constraints from below come in.

Constraints from the Mind

The first set of constraints comes from human cognition. We count in base ten, and the most common explanation is anatomical: we have ten fingers, and early counting used them. Other civilizations made other choices. The Babylonians used a base-sixty system, and the Maya used a base-twenty system. These are real examples of the alternative paths I imagined in high school, and one of them still shapes my daily life: an hour has sixty minutes and a circle has 360 degrees because of the Babylonian choice.

Representations also affect the minds that learn them. In Chinese, number names follow the base-ten structure exactly: eleven is 十一 ("ten-one"), twelve is 十二 ("ten-two"), and twenty-one is 二十一 ("two-ten-one"). English hides this structure behind words like "eleven," "twelve," and "twenty." Miller et al. (1995) found that Chinese-speaking preschoolers had an advantage over English-speaking children in learning to count, while the two groups did not differ on other early skills, such as counting small sets. The difference was not in children's mathematical ability but in how transparently their language represented the base-ten system. I learned to count in Chinese and learned English later as a second language, so I never had to discover that "twelve" means "ten and two." My first number words already told me. The representation did part of the computation for me.

Cognitive constraints can also push different civilizations toward similar representations. Place-value notation, in which a digit's value depends on its position, appears to have developed in several places, including Mesopotamia, China, India, and Mesoamerica, and a symbol for zero was developed independently at least twice, by the Maya and in India. If notation were purely arbitrary, this kind of repeated invention would be surprising. It is less surprising if human minds, with limited memory and a need to calculate quickly, face the same pressures everywhere. A system that lets a few symbols express any number, and lets calculation proceed column by column, is simply easier for a human mind to use.

So the mind constrains representation in two directions at once. It introduces arbitrary features, such as the number ten, and it favours certain designs, such as place value, that different cultures reach separately. Part of the answer to "why this form?" is that this form fits the kind of mind that had to use it.

Constraints from Earlier Representations

The second set of constraints comes from representations that already exist. Once a representation is widely used, later work is built in it, and switching becomes costly; economists call this path dependence (David, 1985). The sixty-minute hour is an example. No one designing a new system of time today would choose sixty, but the old representation is built into everything that came after it.

Path dependence can also shape what gets discovered. Newton and Leibniz developed calculus independently, with different notations. British mathematicians largely kept Newton's dot notation, while mathematicians on the Continent adopted Leibniz's *dy/dx*, which makes the structure of operations such as the chain rule visible on the page. Historians often link Britain's slower progress in analysis during the eighteenth century partly to this choice. In 1812, a group of Cambridge students including Charles Babbage, who later designed some of the first mechanical computers, formed the Analytical Society to promote Leibniz's notation. Babbage later joked that its book should have been titled "The Principles of pure D-ism in opposition to the Dot-age of the University" (Babbage, 1864). The mathematics was the same on both sides. The representation decided what was easy to do with it, and therefore what got done.

This is the recursive process I sensed in high school. Representations are used to compute; the results of computation are written down as new representations; those become the base for the next round. Tomasello (1999) calls this the ratchet effect: humans keep and build on the modifications made by earlier generations instead of reinventing them, so culture accumulates rather than starting over (see also Tennie et al., 2009). A ratchet only moves forward, which is its strength. But it also means that each new representation inherits the choices built into the old ones. When I learn calculus today, I am also learning, without noticing, the decisions of people who chose its notation three centuries ago.

So the constraints from below are not only the limits of the human brain. They include the whole stack of earlier representations underneath any new idea. Each iteration is free to add something new, but only on top of what is already there. Over many iterations, this narrows the range of paths that remain open, and the process converges toward the civilization we now have.

Would Another Civilization Converge?

Putting these constraints together gives a more precise answer to the question I asked about other civilizations, or other universes. At the level of content, I expect strong convergence. Any civilization that needs to count harvests, predict seasons, or build stable structures faces the same problems, and the world will push its answers toward the same relationships. The Greek and Chinese traditions reached the same theorem about right triangles, apparently without depending on each other. At the level of representation, I expect much less convergence. The base, the symbols, which concepts are named first, and what counts as a proof all depend on cognitive accidents, such as having ten fingers, and on historical accidents, such as which notation a country's universities happened to adopt.

One objection is that this distinction is too clean. Perhaps some representations are not just convenient but correct. If mathematical truths exist independently of us, as many mathematicians believe, then a good enough representation might be forced on anyone who studies them long enough, and a sufficiently advanced civilization would end up with something very close to ours.

I think this objection is partly right. The repeated invention of place value and zero shows that some representational choices are driven by deep pressures, not accidents. But convergence on design principles is not convergence on form. Babylonian, Mayan, and Indian place-value systems all used position, yet they differed in base, symbols, and how they marked an empty place. Even within one civilization, the same physics can be written in quite different forms: Newtonian, Lagrangian, and Hamiltonian mechanics describe the same motions, and physicists choose among them depending on the problem. Whether or not mathematical truths exist independently, which is a question I cannot settle here, a single truth can still be represented in many ways. What converges is the set of problems a representation must handle, not the representation itself.

The same reasoning may extend to the social structures I wondered about in high school. Laws and institutions are also shared representations of how people should act toward one another. Recurring problems, such as sharing resources or settling disputes, may push societies toward similar functions, while their particular forms are inherited from earlier forms. I cannot defend this fully here, but the answer has the same shape: different in form, much less different in the problems solved.

A Connected World

This framework also suggests something about the future. Much of the evidence I have used depends on isolation. We know that place value and zero reflect deep pressures because they were invented more than once, and we know the right-triangle theorem is not just a Greek idea because a Chinese tradition reached it apparently on its own. Isolated civilizations worked like separate runs of the same experiment, and comparing their results is how we can tell what is forced by the world from what is an accident of history.

The internet, and now AI, are ending that experiment. Lazer and Friedman (2007) simulated groups searching for solutions to a complex problem and found that more efficiently connected networks did better in the short run but worse in the long run, because fast sharing led everyone to copy the same good-enough solution and removed the diversity needed for further search. A fully connected world may behave the same way: faster progress on shared problems, but fewer independent paths, and so fewer chances to find a representation no one has imagined yet.

AI may strengthen this pull. Doshi and Hauser (2024) found that writers who received story ideas from a language model wrote stories rated as more creative, yet stories written with AI help were more similar to each other than stories written without it. Each individual gained, while the collection lost variety. If future models are trained on text produced by earlier models, the narrowing can feed on itself. Shumailov et al. (2024) showed that models trained recursively on generated data lose the tails of the original distribution, so rare cases disappear first. This is the same recursive process I described above, but with a new risk: when each round is built mainly from the output of the previous one, what is rare may vanish instead of accumulating.

I do not think this means civilization will become uniform. Local languages, problems, and institutions still constrain representations from below, and people can choose to keep alternatives alive. But it does mean that convergence may increasingly come from sharing rather than from the world. In the past, when two isolated civilizations reached the same answer, that agreement was evidence that the answer was forced. In a connected world, agreement will be weaker evidence, because everyone may simply have learned from the same source.

Why We Have to Learn This

The other part of my high-school question was more personal: why do we have to learn things we are not interested in? This analysis gives an answer I did not have at the time. Learning a representation is a way of inheriting computation. When I learned place value, algebraic notation, or calculus, I was not just memorizing arbitrary rules. I was receiving tools that many earlier people had built so that I would not have to solve their problems again. That is the ratchet working as it should.

But it also explains why it rarely felt that way. Representations arrive in school without the problems they were created to solve. A student sees *dy/dx* but not the problems that made Leibniz's notation worth fighting over. Without the problem, a representation looks like an arbitrary rule handed down for no reason, and "why do we have to learn this?" is a reasonable question. In Marr's terms, school often teaches the algorithmic level without the computational level: the steps, without the question they answer.

Looking back, I think my high-school question was less a trap than a mix-up of levels. I was asking why the truth appears in this form, and the answer is that the form does not come from the truth alone. It comes from the minds that had to use it and from the representations that came before. I was asking whether our knowledge could have been different, and the answer is yes, in form much more than in content. And I was asking why I had to learn it, and the answer is that it was the fastest way to stand on the work of everyone before me, even if no one showed me what that work had been for.

If another civilization had started differently, its students would probably be learning different symbols. I suspect they would still be asking the same question.

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